DE–8605 **11**

DISTANCE EDUCATION

M.Sc. (Zoology) DEGREE EXAMINATION, MAY 2014.

ANIMAL DIVERSITY

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions in about 300 words each.

1. Differentiate the prokaryotic and eukaryotic cells.

2. Discuss about the locomotive behavior of protozoan.

3. Barnacles are very successful arthropods. What factors do you think are responsible for the evolution of their highly modified body form?

4. What problems are associated with living and reproducing in terrestrial environments? Explain how insects overcome these problems.

5. What is pentaradial symmetry and why is it adaptive for echinoderms?

6. How are the skeletal and muscular systems of amphibians adapted for life on land?

7. Discuss about the phylogenetic relationships of Hemichordata and Chordata.

8. Write short notes on adaptive features of deep sea animals.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions in about 500 words each.

9. Give a detailed account on structure and functions of cell membranes of animals.

10. Give a detailed account on phylogeny of invertebrates.

11. Briefly explain the following:

    (a) Pseudocoelomate animals.

    (b) Urochordates

    (c) Bony fishes.

12. Briefly explain the general of features of corals and their ecological relationships.

13. Briefly explain the origin and adaptation of birds.

14. Give a detailed account on freshwater fishes and their adaptations.

15. Write short notes on important characteristics of mollusca and their adaptive responses.

DE–8606 **12**

DISTANCE EDUCATION

M.Sc. (Zoology) DEGREE EXAMINATION, MAY 2014.

CELL AND MOLECULAR BIOLOGY

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)
Answer any FIVE questions.
All questions carry equal marks.

1. Write short notes on nucleus and its functions.
2. Comment on properties of prokaryotic DNA polymerase.
3. Add note on steroid receptor.
4. Give an account on post mRNA modification.
5. What are the functions of transcriptional factor?
6. Differentiate inducer and repressor with example.
7. Write short notes on cadherin and autophagy.
8. Write short notes on DNA binding motifs in prokaryotes.

PART B — (4 × 15 = 60 marks)
Answer any FOUR questions.
All questions carry equal marks.

9. Describe the ultrastructure and functions of a Golgi apparatus.

10. Add note on the following
    (a) RNA splicing
    (b) Operon
    (c) Plasmodesmata.

11. Compare and differentiate the organization of Prokaryotic and Eukaryotic Cells.

12. Discuss the trp operon concept.

13. How do protein synthesis intiates, elongate and translocate in prokaryotes?

14. Discuss the promoters, enhancers and process of eukaryotic transcription.

15. In what ways do hormones affect the expression of a gene?

DE–8607 **13**

DISTANCE EDUCATION

M.Sc. (Zoology) DEGREE EXAMINATION, MAY 2014.

GENETICS AND EVOLUTION

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions, each in about 300 words.

1. Write short notes on Barr bodies.

2. Add notes on Hardy Weinberg law.

3. Give an account on Inborn errors of metabolism.

4. Briefly explain the species concept.

5. Explain the structure of gene.

6. Give an account on adaptation pattern and its significance in evolution.

7. Write short notes on clustal analysis.

8. Explain the practical application of ploidy.

PART B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions, each in about 500 words.

9. Give an account of linkage and crossing over.

10. Briefly explain the types of gene mutation and its significance.

11. Explain the role of gene regulation in development and differentiation with example.

12. What is speciation? Explain about allopatric and sympatric speciations.

13. Differentiate between Mendalian inheritance verses Cytoplasmic inheritance.

14. Explain the genetic structure of population. Explain in short about phenotypic variation.

15. Write notes on any three of the following:

    (a) Gene-pool

    (b) Micro evolution

    (c) Isolating mechanism

    (d) Phylogenetic tree.

DE–8608 **14**

DISTANCE EDUCATION

M.Sc. (Zoology) DEGREE EXAMINATION, MAY 2014.

BIOCHEMISTRY AND ANIMAL PHYSIOLOGY

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

All questions carry equal marks.

1. Describe the biological importance of proteins.

2. How vitamins are classified? Explain it with at least two examples.

3. Describe 'obesity' and explain its causes.

4. Describe in detail about the mechanism of digestion of carbohydrate and protein.

5. Explain the gluconeogenesis.

6. Explain the mechanism of formation of urine.

7. Give an account on the structure of nucleic acids.

8. How nerve signals are transmitted?

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following.

All questions carry equal marks.

9. Describe the mechanism of glycolysis? And how it is distinguished from gluconeogenesis?

10. Briefly describe the ultra structure of skeletal muscle.

11. Explain in detail about the types of receptors.

12. Describe the mechanism of glycogen metabolism.

13. What is circadian rhythm? Add notes on the role of melatonin in circadian rhythm.

14. Describe how animals manage temperature stress (thermoregulation).

15. Describe the mechanism of respiration.

DE–8609 **15**

DISTANCE EDUCATION

M.Sc. (Zoology) DEGREE EXAMINATION, MAY 2014.

BIOPHYSICS AND BIOSTATISTICS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

All questions carry equal marks.

1. Explain the basic principles of autoradiography.

2. Describe the principle of Thin Layer Chromatography.

3. Write short notes on Iso Electric Focusing (IEF).

4. Write short notes on NMR spectroscopy.

5. Describe Agarose Gel Electrophoresis.

6. Write short notes on Giegor Muller counter.

7. Describe in detail about the types of variables.

8. Write short notes on the application of ANOVA.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions.

All questions carry equal marks.

9. Briefly explain the basic principle and procedure of NATIVE PAGE.

10. Briefly explain the types of radiations and their impacts.

11. Write about western and southern blotting techniques.

12. Explain in detail about the principle, procedure and application of HPLC.

13. Illustrate and explain the construction of histogram and a polygon and pie diagram with two examples assuming a data.

14. Obtain the mean and standard deviation for the data given.

Marks obtained (Xi) :	10-20	20-30	30-40	40-50	50-60
Number of students (f) :	12	14	23	15	5

15. Explain the significance of Regression and variations in experimental research.

DE–8610 **21**

DISTANCE EDUCATION

M.Sc. (Zoology) DEGREE EXAMINATION, MAY 2014.

ENVIRONMENTAL BIOLOGY

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

All questions carry equal marks.

1. Write short notes on Acid rain.

2. Discuss briefly the energy input in an ecosystem.

3. What is limiting factors? Explain the Shelford's law of tolerance.

4. Write about the trophic structure and their role in ecology.

5. Write short notes on ecological succession.

6. Write a note on population density.

7. What are coral reefs? Write the unique features of coral reefs.

8. What is pollution? Explain the effects of water pollution on the biological system.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

All questions carry equal marks.

9. Write in detail about the energy flow in ecosystem.

10. What is ecological complexity? Explain about its stability in food web.

11. Comment on the following :

    (a) Mortality

    (b) Climax

    (c) Monoclimax

    (d) Population density.

12. Describe about the conservation methods of natural resources.

13. Write about the factors that regulate the population growth.

14. What is bioremediation? Write in detail the role of microbes in bioremediation.

15. What is climatic change? How does the global warming affect the living organisms?

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DE–8611 **22**

DISTANCE EDUCATION

M.Sc. (Zoology) DEGREE EXAMINATION, MAY 2014.

DEVELOPMENTAL BIOLOGY

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

All questions carry equal marks.

1. Explain various stages of spermatogenesis.

2. Describe the process of egg maturation from any one chosen animal model.

3. How do sperms recognize species – specific eggs?

4. Give an account on the types of cleavage with suitable examples.

5. Elaborate importance of organizer in embryogenesis.

6. How do germ cells accomplish determination and cell migration during development?

7. Discuss the sequential stages in formation of neural crest.

8. Explain the factors inducing teratogenesis.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Describe the structure and physiological aspects of spermatozoa.

10. What is vitellogenesis? Explain this process with respect to source of the basic biochemical components.

11. Discuss various factors affecting cleavage.

12. Describe in detail the location and features of presumptive organ forming areas in frog.

13. Elucidate the process of gastrulation in chick with neatly labelled diagrams.

14. Explain the sequential stages in eye development.

15. "Nuclear transplantation studies have significantly contributed to a better understanding of genes in development" – Justify.

DE–8612 **23**

DISTANCE EDUCATION

M.Sc. (Zoology) DEGREE EXAMINATION, MAY 2014.

MICROBIOLOGY AND IMMUNOLOGY

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

All questions carry equal marks.

1. Write briefly the biological importance of microalgae.

2. Define the following terms :
   (a) Capsule
   (b) Spores.

3. Explain the reasons contributing for the stationary and decline phases of bacterial growth.

4. What is a bacterial culture medium? Write the uses of selective and differential media.

5. Write any two methods used to detect the microbes in milk.

6. Write short notes on food poisoning.

7. Describe briefly about the structure of antibody.

8. Write about the various types of antigen.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

All questions carry equal marks.

9. Write in detail the structure of all membrane and cell wall of gram positive bacteria.

10. Explain the various biochemical tests used for identification of microbes.

11. Describe about the structure and functions of primary lymphoid organs.

12. Write in detail the alternative complement system.

13. What is allergen? Explain about the type I hypersensitivity.

14. Write the principle, procedure and clinical significance of immuno electrophoresis.

15. Describe in detail the autoimmune disorders.

DE–8613 **24**

DISTANCE EDUCATION

M.Sc. (Zoology) DEGREE EXAMINATION, MAY 2014.

ANIMAL BIOTECHNOLOGY

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions, each in about 300 words.

1. What is the role of plasmid in genetic engineering? Explain briefly.

2. What is human genome project? Write its potential applications.

3. Write a note on various methods involved in animal cell culture.

4. What are cell lines? How is the growth characteristics of cell lines determined?

5. List out the application of vermiculture.

6. Give an account on vaccine production using biotechnological approaches.

7. What is the scope of bioinformatics? Enlist few of its applications.

8. Write a note on Multiple Sequence Alignment in phylogentic analysis.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions, each in about 500 words.

9. Describe in detail the types of PCR and their applications.

10. Write an essay on the technique of animal cell culture.

11. Describe in detail the bioreactors and scaling-up technologies involved in cell cultures in product development.

12. Describe the life cycle and rearing methods of silkworm, *Bombyx Mori*.

13. What is gene therapy? Explain the limitations and prospects of its use in medicine.

14. Describe in detail various types of bioinformatics tools involved in retrieving literature collection.

15. Discuss in detail the various criteria used in designing primers for PCR.

DE–8614 **251**

DISTANCE EDUCATION

M.Sc. (Zoology) DEGREE EXAMINATION, MAY 2014.

FISHERIES AND AQUACULTURE

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Give a brief account on general classification of fishes.

2. Describe the methods of age determination in fishes.

3. Comment on the soil characteristics in pond construction.

4. Write a short note on live feed organisms and theft culture techniques.

5. Write an account on Paddy cum fish culture. Add a note on problems and prospects of Paddy cum fish culture.

6. Briefly explain the suitable areas for sea weed culture.

7. What is quality assurance? Add a note on HACCP concepts and principles.

8. Give a brief account on the present status of aquaculture in India.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions.

9. Write an account on the economically important freshwater and marine fishes in India.

10. Give a detailed account on food and feeding habits of brackish water fishes.

11. Explain different types of aquaculture practices in India.

12. Write an essay on symptoms and control of fungal and bacterial diseases in fish.

13. Write a detailed account on the culture techniques of pearl. Comment on their economic importance.

14. Define hybridization. Explain the advantages and disadvantages of Induced breeding in India.

15. Define the principles of preservation. Explain in detail the process and methods of preservation.

DE–8615 **252**

DISTANCE EDUCATION

M.Sc. (Zoology) DEGREE EXAMINATION, MAY 2014.

PARASITOLOGY

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

All questions carry equal marks.

1. Give an account on parasitism.
2. What are the different disciplines of medical parasitology?
3. Explain the common pathways for the transmission of parasites.
4. Enlist and explain the six main morphologies of trypanosomatids.
5. Give a detailed account on Cyclops.
6. Describe the integrated program for controlling mites.
7. Discuss the various strategies of vector control measures.
8. Give an account on Bioterrorism.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

All questions carry equal marks.

9. Discuss the life cycle of Schistosomiasis.
10. Describe the molecular lab diagnosis and its significance in parasitology.

11. Discuss the Diagnosis, treatment and prevention for Lymphatic filariasis.

12. Explain the role of different Mosquito sp in disease spread.

13. Give a detailed account on Anti malarial agents based on their mechanism of action.

14. Explain the life cycle of *Entomoeba histolytica* with suitable illustration.

15. Describe the characteristic properties of Ascaris lumbricoides.